A/B Testing & Statistical Analysis



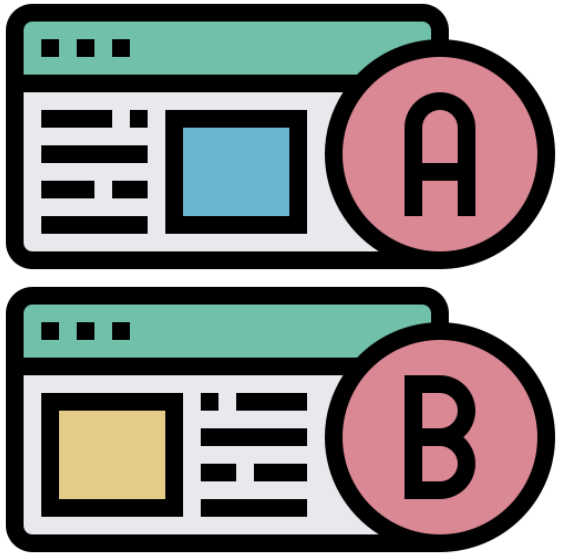


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Topics

1. A/B Testing Fundamentals
2. Use Cases Across Industries
3. How A/B test works
4. Experiment flow & Terminology
5. Errors in A/B testing
6. Statistical methods for analysis
7. Real-world A/B test example with analysis
8. **Special Bonus**



1. What is A/B Testing



A randomized method of comparing two or more versions to see which one performs better. It helps in making data-driven decisions.

- **Control group** (version - A): Existing version
- **Treatment group** (version - B): New version to test

Types of A/B Tests

1. Standard A/B Tests – Compare two versions (A and B), for one variable.
2. A/B/n Tests – Compare multiple versions (A, B, C, etc), for one variable.
3. Multivariate Test (MVT) – Compare multiple variables simultaneously.

Quick Example of A/B Testing

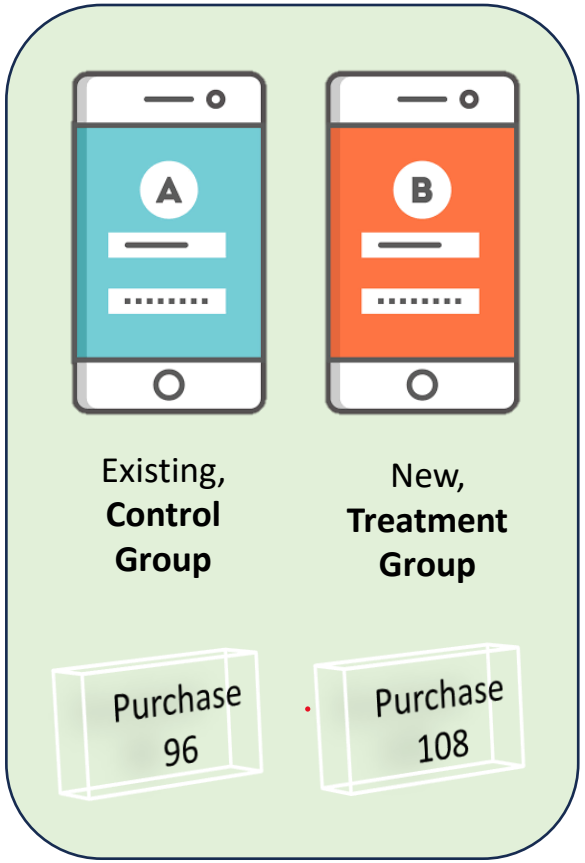
A company wants more people to buy products, so data analytics team has an idea: *"What if we change the background color of the checkout page from blue to orange?"*

Instead of changing it for everyone immediately, we conduct an experiment.

- **Randomly divide users into two groups**: Group A sees existing version (blue color, control group), and Group B sees new version (orange color, treatment group).
- **Measure the results**: Compare how many users in each group actually purchase a product.

If Group B generates more purchases, the new orange color might be better.

However, we must check whether the difference is **statistically significant** before concluding that the new design truly helped.



2. A/B Test Use Cases Across Industries



1. Product and Website

- CTA button text (Pri metric: Conversion rate), Simplified onboarding page designs (Pri metric: Signup completion rate)

2. Marketing and Advertising

- Email subject lines (Pri metric: CTR), adv creatives or landing page designs (Pri metric: conversion rate)

3. E-commerce and Retail

- Free-shipping thresholds (Pri metric: Profit per visitor), discount offers (Pri metric: Purchase conversion rate)

4. Subscription and SaaS

- Monthly and annual pricing page layouts (Pri metric: Paid subscription conversion rate), 7-day free trial against a 14-day free trial (Pri metric: Free-to-paid conversion rate)

5. Mobile Apps and Gaming

- Different game tutorials to help new users understand the gameplay (Pri metric: Tutorial completion rate), Push notification messages to improve user engagement (Pri metric: 7-day retention rate).

6. Search Engines and Recommendation Platforms

- Search-ranking algorithms to improve search relevance (Pri metric: Successful search rate), Personalized recommendations with generic recommendations (Pri metric: Click-through rate).

3. How does an A/B test work?

As a Data Analyst, you will be responsible for defining metrics, validating the data, writing SQL queries, calculating results, interpreting statistical tests, and presenting recommendations to stakeholders.

Step 1: Define the business problem

Example: Increase the purchase conversion rate.

Step 2: Create a hypothesis

The new checkout page will increase purchases.

Step 3: Randomly assign users

Group A gets the old page; Group B gets the new page. Setup experiment.

Step 4: Collect and analyze data

Calculate conversion rates, statistical analysis and compare result.

Step 5: Make a decision

Launch the change, reject it, or investigate further based on the evidence.

4. Experiment Terminology



1. **Control group (A):** the group that receives existing version. It acts as baseline for comparison.
2. **Treatment group / Variant (B):** the group that receives new experience you want to test.
3. **Hypothesis:** A statement you want to test. For example, conversion-rate:
   - **Null** hypothesis (H_0): There is no difference in the true conversion rates of Group A and Group B.
   - **Alternative** hypothesis (H_1): There is a difference in their true conversion rates.
4. **Minimum Detectable Effect (MDE):** is the smallest difference your experiment is designed to reliably detect at a specified significance level and power.

   Why is MDE important? A small improvement is generally harder to detect than a large one, so it typically requires a larger sample size.

Experiment Terminology

5. **Primary metric:** The main outcome used to judge the experiment's success, such as purchase conversion rate.
6. **Guardrail metrics:** Metrics that should not get worse while improving the primary metric. For eg, average order value, refund rate, page load time or complaint rate.
7. **p-value:** probability of observing a difference between control and treatment. If,

   p-value =< 0.05, result is statistically significant,

   p-value > 0.05, result is not statistically significant.
8. **Statistical significance:** Evidence that the observed difference is unlikely to be explained by random variation alone, under the assumptions of the statistical test
9. **Significance Level** (Alpha / α)**:** Predetermined risk threshold for a false positive (**type I error**), most commonly set at **0.05**. And (1- α) is **confidence level,** i.e. **95%**.

Experiment Terminology

10. **Statistical power:** the probability that your experiment detects an effect of a specified size when that effect really exists.

    The formula is: $\text{Power} = 1 - \beta$

    If you choose 80% power: $\beta = 1 - 0.80 = 0.20$

    This means the experiment has an 80% chance of detecting the specified effect, assuming the effect exists and the design assumptions hold. There is a **20% Type II** error probability for that effect.

    Why is power important?

    Higher power reduces the risk of missing a meaningful improvement. The usual trade-off is that higher power requires a larger sample size.

Other terms worth knowing

- **Randomization**: Assigning users to groups randomly to reduce selection bias and make the groups comparable.
- **Sample Ratio Mismatch (SRM):** When the observed allocation between groups differs unexpectedly from the planned allocation—for example, you expected a 50/50 split but observed 60/40. Investigate tracking, assignment logic, eligibility rules, or data pipelines before trusting the results.
- **Peeking**: Repeatedly checking results and stopping the experiment when a favourable p-value appears. This can inflate false-positive rates if the analysis does not account for repeated testing.
- **Standard error**: measures the uncertainty in a statistic caused by sampling variation.
  - When we run an A/B test, we don't usually test every single user in the world- we test a sample (a slice) of your audience.

5. Errors in Experiment



1. **Type I Error (False Positive)**

   Type I error happens when you conclude that the new version works, even though there is no real improvement.

   Example, imagine the new checkout is actually no better than the old one. But your experiment produces a result that looks statistically significant, so you launch the new page.

   To avoid this, we set the significance level (**α**) to 5% (0.05) to control the long-run false-positive rate. So, the test has a 5% Type I error rate under the null hypothesis.

   Simple words, "You launch a change that has no true effect."

2. **Type II Error (False Negative)**

   Type II error happens when a real improvement exists, but your experiment fails to detect it.

   Example, suppose the new checkout genuinely improves conversion from 5% to 6%, but your sample is too small. The test returns a non-significant result, so you conclude that there is insufficient evidence of an improvement. You may miss a valuable business opportunity.

   This is a Type II error, represented by 'beta' β.

   Simple words, "You miss a real improvement."

How sample size, MDE, Error & Power Connect

1. Define the experiment
   Baseline conversion rate and MDE
2. Choose statistical settings
   Significance level α and power $1 - \beta$ (type I and II error)
3. Calculate required sample size & days to run
   Determine how many users each group needs *(minimum users)*
4. Run and analyze the experiment
   Estimate the effect and evaluate the statistical evidence

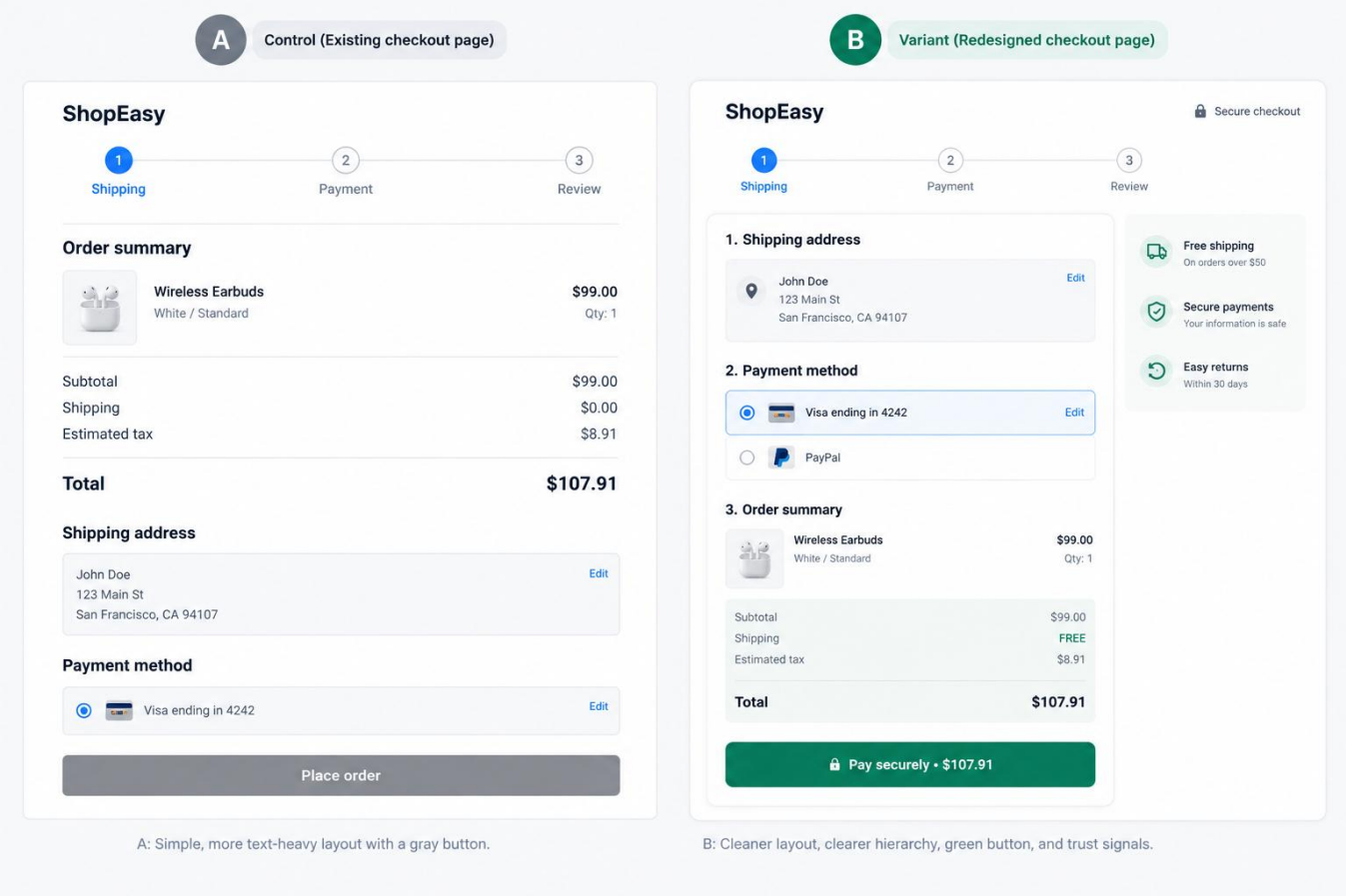
6. Statistical Methods to Analyze Test Results



- **Z-test:** Checks whether the **ratios** of two groups with **large sample** sizes differ significantly. Z-test calculates, **Z-score** which tells how far a value is from the mean, measured in standard deviations (critical value of approximately **1.96**). Z-test is for percentages or proportions.
  - Examples: Conversion rate, Click-Through-Rate (CTR), etc.
- **T-test:** Checks whether the **means** of two groups differ significantly. T-test is used for averages or continuous metrics (numbers).
  - Examples: Average Order Value (AOV), revenue per user, session duration, etc.
- **Chi-Square test:** Checks whether **categorical variables** are associated to each other or their proportions differ from what expected. Used when data is split into groups or categories.
  - Examples: Multi-Option Preference Survey (four different **app color** themes to see if preferences vary by user demographic), Browser/Device Distribution Check (distribution of users is equal and random across all **devices-** mobile, tab and desktop), etc.

7. Purchase Conversion Rate Experiment

Business scenario: An e-commerce company wants to increase purchases by redesigning its checkout page.



Purchase Conversion Rate Experiment

Step 1. Define the objective and hypothesis

- Objective: Increase purchase conversion rate.
- Control (A): Existing checkout page.
- Variant (B): Redesigned checkout page.
- Primary metric: Purchase conversion rate = users who purchase ÷ eligible users exposed to the checkout page.
- Guardrail metrics: Average order value, payment failure rate, checkout errors and complaint rate.

Hypotheses:

- Null hypothesis (H0)): The redesigned checkout page does not change the true purchase conversion rate compared with the existing checkout page. H0, Conversion rate B = conversion rate A.
- Alternative hypothesis (H1): The redesigned checkout page changes the true purchase conversion rate compared with the existing checkout page. H1, Conversion rate B ≠ conversion rate A.

Purchase Conversion Rate Experiment

Step 2. Set experiment parameters

Assume historical data shows a 5% purchase conversion rate.

Parameter	Value
Baseline conversion	5%
MDE	1 percentage point (5% → 6%)
Significance level (α)	5%
Statistical power	80%
Allocation	50% A / 50% B

Using these assumptions, a two-proportion **sample-size** calculation requires approximately 8,160 users per group (about 16,320 total) – minimum target users.

To calculate sample size, we can use any 3rd party tools, else can use formula and do it manually.

Purchase Conversion Rate Experiment

Step 3. Run the experiment

- Randomly assign eligible users 50/50 to A or B.
- Keep each user's assigned version consistent.
- Run until the planned sample size and experiment duration are met.
- Validate tracking, randomization, and sample-ratio balance.
- Avoid stopping early just because results look favourable.

Step 4. Analyze the results

Suppose the experiment completes with these results:

Metric	A: Existing	B: Redesigned
Users	10,000	10,000
Purchases	500	600
Conversion rate	5%	6%

Purchase Conversion Rate Experiment

Calculate the impact:

- Absolute lift: 6% − 5% = 1 percentage point
- Relative lift: (6% − 5%) ÷ 5% = 20%
- Statistical test: Two-proportion z-test (z-score: 3.1)
- p-value: Approximately 0.002
- 95% confidence interval for the lift: Approximately +0.63 to +1.37 percentage points.
    - Since it's entirely above zero, so the data support a positive improvement.

Since p-value (0.002) is below α (0.05), & entire confidence interval was above zero, so result is **statistically significant.**

Step 5. Interpret and recommend

- Result: Group-B performed better.
- Conversion improved by 20% relative to A.
- The observed improvement is statistically significant.
- Check guardrail metrics to ensure the redesign did not harm order value or payment success.
- If the results and business impact remain favourable, recommend a staged **rollout** and continued monitoring.

Purchase Conversion Rate Experiment

Calculate the Lift for 95% confidence interval:

- **Absolute lift**: 6% − 5% = 1 percentage point
- 95% confidence interval for the lift: Approx +0.63 to +1.37 percentage points.
  - Since it's entirely above zero, so the data support a positive improvement. It takes your single lift number (**1 pp**) and wraps a safe boundary around it to account for uncertainty

$$\text{Confidence Interval} = \text{Absolute Lift} \pm (1.96 \times \text{Standard Error})$$

$$\text{Margin of Error} =$$

$$\text{Margin of Error} = \text{roughly } 0.37\%$$

$$\text{Lower bound} = 1.00\% - 0.37\% = +\mathbf{0.63\%}$$

$$\text{Upper bound} = 1.00\% + 0.37\% = +\mathbf{1.37\%}$$

Formula: Standard Error (The Uncertainty)

$$SE = \sqrt{\frac{p_A(1-p_A)}{n_A} + \frac{p_B(1-p_B)}{n_B}}$$

(Where p is your conversion rate and n is the number of users).

*To create a 95% confidence interval, statisticians take the exp's center point (**1 pp**) and add/subtract standard error multiplied by 1.96 (the magic number for 95% confidence).*

Bonus: Add A/B Testing as Skill in Resume Examples



1. Product and Website - Checkout Page Redesign

Analyzed checkout funnel drop-offs and user behavior, partnered with product and engineering teams to optimize the checkout experience, and evaluated results through A/B testing, increasing checkout conversion rate by 12% and reducing cart abandonment by 8%.

2. Marketing and Advertising - Landing Page Optimization

Evaluated landing page performance across traffic sources and audience segments, conducted A/B tests on page layouts and CTAs, and identified high-performing designs that increased landing page conversion rate by 15% and reduced cost per acquisition (CPA) by 10%.

3. E-commerce and Retail - Discount Offer Optimization

Analyzed customer purchase behavior and discount performance across customer segments, evaluated promotional strategies through A/B testing, and identified optimal discount offers that increased purchase conversion rate by 10% while improving average order value (AOV) by 5%.

Bonus: How to Explain in a Job Interview



Use the **STAR Method** (Situation, Task, Action, Result) to frame your response. 🚀

1. Situation & Task (The Setup): I built an end-to-end A/B testing project analyzing a redesigned conversion funnel. The goal was to determine if a new checkout design improved purchase conversions compared to the existing baseline.

2. Action (The Analytical Rigor)

- I evaluated a sample of 20,000 users split evenly between Control (A) and Treatment (B). The baseline conversion rate was 5%, and the treatment achieved 6%.
- To ensure this wasn't just sampling variation or random chance, I calculated the standard error and constructed a 95% confidence interval for the absolute lift. The absolute lift was **+1.00 percentage point**, with a 95% confidence interval ranging from **+0.63 to +1.37 percentage points** (or a **+20% relative lift**).
- Additional step: I also ran a **Chi-Square test** to check whether user distributions across devices (mobile, tab and desktop) shifted significantly between variants.

3. Result & Business Takeaway (The Decision): Because for this current setup, p-value is less than 0.05 and the entire confidence interval was above zero, I confirmed that the result was statistically significant and recommended rolling out Version B to 100% of traffic, projecting an overall increase in purchase volume.

Interview Tips 🎯

- **Don't fake real traffic:** Say "This was a project I built to simulate an enterprise-scale experimentation pipeline.“.
- Be prepared for the Follow-Up Questions. And remember, Interviewers care far more about how cleanly you think about experimentation than whether the website had real users or not.

RESOURCES

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